

Project 4 Central Bank Communications

Central Bank Communications Corpus

Collection methods and documentation for 302 communications

BRICS Trade, Labour & Structural Change Portfolio

Tool: Python (requests, BeautifulSoup, Selenium, pandas)

Output: data/brics_mpc_statements_v2.csv

Feeds into: Project 5 — Monetary Policy Sentiment Analysis

1 Background and Motivation

The BRICS economies — Brazil, Russia, India, China, and South Africa — collectively represent a significant and growing share of global trade, output, and financial flows. Despite sharing a geopolitical label, their monetary policy frameworks, communication styles, and responses to shared macroeconomic shocks diverge considerably.

Since the 2008 Global Financial Crisis, central bank communication has been recognised as a monetary policy instrument in its own right. Forward guidance, the tone of post-meeting statements, and the framing of inflation and growth outlooks all affect market expectations and the transmission of monetary policy — particularly in emerging markets where credibility and communication clarity matter more, not less.

For BRICS economies, this communication dimension is especially interesting given:

- Currency fragmentation pressures and divergent exchange rate regimes
- Differential exposure to global trade shocks, including the US-China trade war and post-COVID supply disruptions
- Varying degrees of central bank independence and inflation targeting credibility
- The evolving geopolitical positioning of BRICS as a bloc with implications for trade invoicing and reserve currencies

This project assembles 302 English-language communications from PBOC, RBI, SARB and CBR. It supplies Project 5 with text for descriptive dictionary sentiment and topic analysis. The sources differ in genre and coverage; PBOC speeches are broader than formal MPC statements, and sentiment is not a calibrated hawkish or dovish policy-stance measure.

2 Research Context

A growing body of academic literature applies natural language processing to central bank communications. Key contributions include:

- Mathur & Sengupta (2019) — quantitative analysis of RBI monetary policy statements from 1998 to 2018, showing a persistent semantic shift following inflation targeting adoption
- Coco & Viegi (2019) — NLP analysis of SARB communications showing a move toward more forward-looking and balanced communication
- BIS Working Paper 1299 (2025) — topic modelling of nearly 20,000 central bank speeches across 100 institutions from 1997 to 2025
- Multiple studies on COPOM communications using LDA and dictionary methods to classify Brazilian central bank communications as hawkish or dovish

The corpus supports comparative text analysis subject to differences in document type, language and time coverage. Those differences must be considered before interpreting bank-level contrasts.

This is directly relevant to the portfolio's overarching theme: trade, structural change, and labour markets in emerging economies. Monetary policy communication is one of the channels through which global trade shocks are transmitted to domestic conditions in these economies.

3 Data Collection Strategy

Four of the five original BRICS central banks are represented in the saved corpus. The counts below refer to unique records in the final v2 snapshot, not successful-retrieval rates or a complete census of each institution.

Country	Institution	Method	Coverage	Documents
India	RBI	HTML plus 57 PDF extracts	2016–2026	63
Russia	CBR	Direct HTTP scrape	2018–2026	51
South Africa	SARB	Selenium browser scrape	2006–2026	57
China	PBOC	BIS CBSPEECHES database	1996–2025	131
Brazil	BCB	Excluded	—	—

3 1 Reserve Bank of India RBI

The RBI publishes a Governor's Statement after each MPC meeting — the formal post-decision communication announcing the rate decision and policy stance. These are archived at rbi.org.in with a clean year-based URL structure (Annualpolicy.aspx?Year=XXXX).

The final RBI corpus has 63 documents dated 5 April 2016 to 6 February 2026: six unique HTML records and 57 historical PDF extracts. The original scraper CSV contains 66 rows because the same six URLs were returned repeatedly.

The scraper (`scrapers/rbi_scraper.py`) loops through each year's archive page, identifies Governor's Statement links using Unicode-aware text matching (the apostrophe in Governor's Statement is a curly Unicode character, not ASCII), fetches each page, and extracts the full text from the `text1` div.

3 2 Bank of Russia CBR

The CBR publishes key rate press releases after each Board of Directors meeting. These are available in English at cbr.ru. The CBR adopted its current key rate framework in September 2013.

The scraper (`scrapers/cbr_scraper.py`) fetches all press release links from the CBR calendar page (cbr.ru/eng/dkp/cal_mp/), which lists decisions going back to 2018 in a consistently structured format. Each individual press release page contains the full statement text in the `landing-text` div. 51 statements were collected from 2018 to 2026.

A methodological note: Bank of Russia statements from 2022 onwards are included but interpreted with awareness of the geopolitical context affecting Russia's external communications and data availability. This is noted transparently in line with standard academic practice.

3 3 South African Reserve Bank SARB

The SARB publishes MPC statements after each Monetary Policy Committee meeting. Statements are available at resbank.co.za going back to 2006, giving us 124 candidate pages.

The historical collection used headless Chrome through Selenium to obtain rendered page content. Live site behaviour may change; a new collection may require updated selectors and access methods.

The SARB scraper visits up to five index pages and joins cmp-text blocks. The saved extract has 57 records. Failed or short pages are omitted; a complete historical failure manifest was not retained, so a verified retrieval success rate is unavailable.

3.4 People's Bank of China PBOC

The PBOC does not publish clean post-meeting press releases in English comparable to the other central banks in this dataset. English-language communications on the PBOC website are sparse and inconsistently structured, with the site relying heavily on JavaScript rendering and offering limited English content.

PBOC material comes from the BIS CBSPEECHES bulk download. The retained records include speeches and broader communications rather than a homogeneous set of formal MPC statements.

The description filter matches People's Bank of China, PBOC or PBC. The saved selection has 131 entries dated 10 September 1996 to 27 October 2025. This filter is not a formal document-type classifier.

3.5 Banco Central do Brasil BCB Excluded

The BCB publishes COPOM statements through an open data API (dadosabertos.bcb.gov.br). However, the API returns statements only in Portuguese. The BCB English website is JavaScript-rendered with no accessible endpoint returning English text.

Machine translation was considered but rejected on the grounds that it would introduce systematic noise into the sentiment analysis in Project 5, where the Loughran-McDonald financial dictionary and other NLP tools are calibrated for English text. Including machine-translated text alongside original English text would create an inconsistent corpus.

Brazil is absent from this collection. The retained corpus covers four institutions with heterogeneous document types and time coverage; it is not a uniform cross-bank MPC panel.

4 Dataset Output

The canonical output is `data/brics_mpc_statements_v2.csv`. The offline builder combines saved bank extracts and `rbi_historical.csv`, retains the first record per bank and URL, and checks required fields, empty text and repeated bank-text combinations.

Column	Description	Example
country	Country name	India
central_bank	Institution abbreviation	RBI
date	Raw source date; precision varies	February 06, 2026
title	Statement title	Governor's Statement: February 06, 2026
text	Full statement text (raw)	Good morning. I welcome you all...
url	Web URL or local PDF reference	https://www.rbi.org.in/Scripts/...

Final snapshot: 302 unique communications from four banks. The older `brics_mpc_statements.csv` has 305 rows, including 60 repeated RBI rows; it is an intermediate file, not the analysis corpus.

5 Technical Summary

The collection combines HTML extraction, browser-rendered pages, BIS bulk text and local RBI PDFs. PyMuPDF extracts page text from 57 PDFs; dates come from filenames and titles are generated from those dates. No OCR is implemented.

Technique	Library	Used For	Key Challenge Overcome
HTTP scraping	requests, BeautifulSoup	RBI, CBR	Unicode apostrophe matching (RBI); calendar page navigation (CBR)
Headless browser scraping	Selenium, webdriver-manager	SARB	Network-level anti-bot block; JavaScript-rendered content
Bulk dataset download + filtering	pandas	PBOC	No clean English API; used BIS pre-compiled corpus
PDF text extraction	PyMuPDF	57 RBI PDFs	Filename dates; no OCR or original URL manifest

The six raw fields are country, central_bank, date, title, text and url. PDF records use local_pdf:<filename>, not an original download URL. The source URL manifest for those PDFs is unavailable. Empty or unreadable documents are skipped with console messages; there is no complete persistent failure log.

6 What Comes Next Project 5

Project 5 applies LM dictionary scoring to all 302 communications, a FinBERT comparison to 40 selected documents, and LDA topic modelling.

Layer 1 — Dictionary-Based Sentiment Analysis

LM measures financial-text polarity and uncertainty using a dictionary developed for corporate filings. It is not calibrated to classify hawkish versus dovish monetary policy. Bank-level differences may reflect source genre, vocabulary and coverage.

The sentiment series are descriptive. Event timing alone does not identify causal responses to trade fragmentation, sanctions or other macroeconomic developments.

Layer 2 — LDA Topic Modelling

LDA is fitted to cleaned text. The reported model has nine topics and coherence 0.505. Topic labels summarise the fitted corpus rather than validated policy categories.

FinBERT is a stratified 40-document comparison, ten per bank. The code uses the first 400 words and tokenizer truncation at 512 tokens. The reported correlation with LM scores is 0.441 ($p = 0.004$) in that sample, not full-corpus validation.

Data Cleaning Before Analysis

Project 5 removes boilerplate, repairs dates and preprocesses the text. The following limitations remain:

- RBI and SARB text fields contain boilerplate header text (press release metadata, repo rate notices) that should be stripped before NLP processing
- Forty-six SARB records have year-only dates, with January placeholders and date_approximate=True. Month-level dates also do not establish exact meeting days.
- PBOC entries span a wider time range (1996–2025) and include a broader range of communication types than the other three banks — subgroup analysis may be warranted

For offline reproduction, run `python 04_central_bank_scraper/build_corpus.py --output /tmp/brics_mpc_rebuilt.csv` from the repository root. All 302 records can be reconstructed from saved extracts. Fresh live scraping can yield different coverage and should retain retrieval and failure logs.

7 Repository Structure

The files below separate canonical outputs, historical intermediates and acquisition sources.

File / Folder	Description
04_central_bank_scraper/notebook.ipynb	Main narrative notebook — runs all scrapers and assembles master CSV
scrapers/rbi_scraper.py	RBI Governor's Statement scraper (HTTP)
scrapers/cbr_scraper.py	CBR key rate press release scraper (HTTP)
scrapers/sarb_scraper.py	SARB MPC statement scraper (Selenium)
scrapers/pboc_scraper.py	PBOC data source documentation (BIS)
scrapers/bcb_scraper.py	BCB exclusion documentation
data/rbi_statements.csv	66 rows representing six unique HTML documents
data/cbr_statements.csv	51 CBR key rate press releases, 2018–2026
data/sarb_statements.csv	57 SARB MPC statements, 2006–2026
data/pboc_statements.csv	131 PBOC speeches/statements, 1996–2025
data/brics_mpc_statements.csv	Historical intermediate with repeated RBI records
data/brics_mpc_statements_v2.csv	Canonical 302-document corpus
data/rbi_historical.csv	57 PDF text extracts
build_corpus.py	Offline deduplication and final assembly